

Education

Princeton University*PhD student, ORFE*

Princeton

2021–now

Advisors: Matias D. Cattaneo, Boris Hanin

Lomonosov Moscow State University*Specialist (=master's) degree, with honors*

Moscow

2014–2020

Faculty of Mechanics and Mathematics, Department of Probability Theory

Professional Experience

Meta Superintelligence Labs*Research Scientist Intern*

Menlo Park

Sep. 2025 — Apr. 2026

Working on model robustness and hyperparameter transfer.

Yandex*Software Engineer Intern → Junior Software Engineer → Software Engineer*

Moscow

Oct. 2018 — Aug. 2021

Worked in the Advertising Network development team, programming in C++ with CUDA and Python (ML and infrastructure code), administered multiple distributed services.

Research

In [2]–[10], authors are listed alphabetically.

- [1] Shigida B. I., Hanin B., Gromov A. *Learning Rate Transfer in Normalized Transformers*. arXiv:2604.27077, 2026.
- [2] Dong Y., Shigida B. I. *When Independent Sampling Outperforms Agentic Reasoning*. arXiv:2605.08478, 2026.
- [3] Cattaneo M. D., Shigida B. I. *The Effect of Mini-Batch Noise on the Implicit Bias of Adam*. arXiv:2602.01642, 2026.
- [4] Cattaneo M. D., Shigida B. I. *Modified Loss of Momentum Gradient Descent: Fine-Grained Analysis* arXiv:2509.08483. Submitted to JMLR, 2025.
- [5] Cattaneo M. D., Feng Y., Shigida B. I. *Uniform Estimation and Inference for Nonparametric Partitioning-Based M-Estimators*. Annals of Statistics (forthcoming), 2026.
- [6] Cattaneo M. D., Shigida B. I. *How Memory in Optimization Algorithms Implicitly Modifies the Loss*. NeurIPS 2025.
- [7] Cattaneo M. D., Klusowski J. M., Shigida B. I. *On the Implicit Bias of Adam*. ICML 2024.
- [8] Bulinskaya E. V., Shigida B. I. *Discrete-Time Model of Company Capital Dynamics with Investment of a Certain Part of Surplus in a Non-risky Asset for a Fixed Period*. Methodology and Computing in Applied Probability, 2021.
- [9] Bulinskaya E. V., Shigida B. I. *Sensitivity Analysis of Some Applied Probability Models*. English version: Journal of Mathematical Sciences, 2021.
- [10] Bulinskaya E. V., Shigida B. I. *Modeling and Asymptotic Analysis of Insurance Company Performance*. Communications in Statistics Part B: Simulation and Computation. 2019.

Teaching & Service

2022–now Teaching assistant for courses at Princeton University: ORF-309 Probability and Stochastic Systems, ORF-245 Fundamentals of Statistics, ORF-524 Statistical Theory and Methods.

Peer review: NeurIPS, Econometric Theory

2021 Teaching assistant for the course “Probabilistic and Statistical Methods in Machine Learning” at the Vega Institute in Moscow

2015–2020 Member of the organizing committee of the Moscow Math Olympiad, multiple stages of the All-Russian Olympiad

2015 Trained top school students in Moscow for the All-Russian Mathematics Olympiad at the Moscow Center for Continuous Mathematical Education

Awards & Competitions

- Associate member of Sigma Xi (2026)
- Gordon Wu Fellowship at Princeton (2021–now)
- MSU Faculty of Mechanics and Mathematics, Partial Differential Equations Contest: 2nd prize (2017)
- Moscow Math Olympiad: 1st prize (2014)
590 participants, top 16 got the 1st prize
- Moscow Physics Olympiad: 1st prize (2014)
3079 participants, top 42 got the 1st prize
- All-Russian Mathematical Olympiad, final stage: honorable mention (2014)
111 invited participants, 35–38th place
- All-Russian Team Programming Olympiad, final stage: 2nd prize (2013)
172 invited teams, 19th place
- All-Ukrainian Mathematical Olympiad, final stage: winner (2013)
Part of the Moscow guest team; the result is equivalent to the 1st prize
- Moscow Math Olympiad: 3rd prize (2013)
323 participants, top 48 got the 1st/2nd/3rd prize
- All-Russian Mathematical Olympiad, regional stage: absolute winner (2013)
360 invited participants in Moscow, 1–6th place
- Moscow Math Olympiad: 3rd prize (2012)
577 participants, top 22 got the 1st/2nd/3rd prize
- All-Russian Mathematical Olympiad, final stage: prize (2012)
106 invited participants, 21–29th place
- All-Russian Mathematical Olympiad, regional stage: absolute winner (2012)
362 invited participants in Moscow, 1–5th place